🗓️ One-Week Wake Word Project Roadmap

Start: Wednesday, July 23 (Today)

Deadline: Wednesday, July 30

Current Status

Metric	Value
Positive samples ("Hello Nova")	39
Negative samples (everything else)	66
Total real samples	105
With augmentation	~370
Latest F1 Score	0.98
Known issue	False positives on certain clips (test5, test6)

Code Upgrades (I Will Do These Now)

Before you start collecting data, I will upgrade all 4 Python files to make the AI significantly smarter:

1. Richer Audio Features — MFCC Deltas

Currently the model only looks at raw MFCC coefficients (13 features). I will add **delta** (velocity) and **delta-delta** (acceleration) features, tripling the audio information to **39 features per frame**. This lets the model understand how sound *changes over time*, not just what it sounds like at one instant.

2. Feature Normalization

Right now, raw MFCC values can range wildly. I will normalize them to zero mean and unit variance, which makes training far more stable and prevents the model from fixating on loud vs quiet samples.

3. More Training Epochs + Learning Rate Scheduling

I will increase epochs from 30 → **80** and add a learning rate scheduler that automatically reduces the learning rate when the model stops improving. This prevents the model from overshooting the optimal solution.

4. Batch Testing Script

I will create a new `src/test_all.py` script that tests every `.wav` file in the `tests/` folder at once, so you can quickly see if your changes helped.

Day-by-Day Plan

📅 Day 1 — Wednesday, July 23 (TODAY)

Goal: Code upgrades + First big data collection session

I will do:

- [x] Upgrade `extract_features.py` with MFCC deltas + normalization

- [x] Upgrade `train.py` with 80 epochs + LR scheduler

- [x] Upgrade `model.py` to handle 39 input features

- [x] Create `src/test_all.py` batch testing script

- [x] Retrain and verify

You will do:

- Record **30 new "Hello Nova" clips** on WhatsApp voice notes

  - 10 clips speaking normally

  - 10 clips speaking softly / from a distance

  - 10 clips speaking fast / with different intonation

- Record **20 new negative clips**

  - 5 clips of silence / room noise

  - 5 clips of similar-sounding words ("Hello", "Nova", "Yellow", "Over")

  - 5 clips of random conversation / TV / music

  - 5 clips of coughing, breathing, door sounds

End of Day 1 Target: 69 positive, 86 negative samples

📅 Day 2 — Thursday, July 24

Goal: Hard Negative Mining + More data

You will do:

1. Run `py src/test_all.py` to see which test clips still fail

2. **Any false positive clips → copy them into `data/dataset/negative/`** (this is Hard Negative Mining)

3. Record **20 more "Hello Nova" clips** (try different rooms, different mic distances)

4. Record **20 more negative clips** (focus on things that sound like "Hello Nova" — similar words, mumbling)

5. Re-extract and retrain:

```powershell
py src/extract_features.py

py src/train.py

py src/test_all.py
```

End of Day 2 Target: 89 positive, 106+ negative samples

📅 Day 3 — Friday, July 25

Goal: Get to 100+ positive samples + Stress test

You will do:

1. Record **20 more "Hello Nova" clips**

   - Try saying it while walking

   - Try saying it with background music playing

   - Try saying it from across the room

2. Record **15 more negatives** (random speech, other names)

3. Retrain: `py src/extract_features.py && py src/train.py`

4. **Stress test:** Record 10 brand new test clips (5 positive, 5 negative) that are NOT in the training data. Put them in `tests/` and run `py src/test_all.py`

5. Any failures → add to training data as hard negatives/positives

End of Day 3 Target: 109+ positive, 121+ negative. F1 should be ≥ 0.95

📅 Day 4 — Saturday, July 26

Goal: Get a friend to test + Diversity push

You will do:

1. Ask **1-2 friends/family** to record 10 "Hello Nova" clips each

   - Different voices make the model much more robust

2. Record **20 more negatives** from real-life scenarios

   - Leave your phone recording for 2 minutes in different rooms and chop the audio into clips

3. Retrain and test

End of Day 4 Target: 130+ positive, 140+ negative

📅 Day 5 — Sunday, July 27

Goal: Polish and fine-tune

You will do:

1. Run `py src/test_all.py` — aim for **zero false positives** and **zero false negatives**

2. Any remaining failures → add to training data

3. Retrain 2-3 times if needed, checking test results each time

4. Try adjusting the confidence threshold (in `predict.py`) up or down to find the sweet spot

5. Document your final test results

End of Day 5 Target: 0 false positives, 0 false negatives on test set

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📅 Day 6 — Monday, July 28

Goal: Final data push + Report preparation

You will do:

1. Record **any remaining clips** to reach the final targets:

  - **150+ positive** samples minimum

  - **150+ negative** samples minimum

2. Final retrain

3. Run `py src/test_all.py` one last time

4. Start preparing your project report / presentation

5. Note down your final Precision, Recall, and F1 scores

📅 Day 7 — Tuesday, July 29

Goal: Final verification + Submission prep

You will do:

1. Record **5 completely fresh test clips** (not used in training)

2. Run predictions on each — verify they all work

3. Clean up any leftover files

4. `git add . && git commit -m "final version" && git push`

5. Finalize your report

🎯 Final Targets

Metric	Target
Positive samples	150+
Negative samples	150+
With augmentation	1000+
F1 Score	≥ 0.97
False positives on test set	0
False negatives on test set	0

Quick Reference Commands

```powershell
```

```
# Always run from D:\voice_recognizer

# Step 1: Extract features (after adding new audio)
py src/extract_features.py

# Step 2: Train the model
py src/train.py

# Step 3: Test ALL clips at once
py src/test_all.py

# Step 4: Test a single clip
py src/predict.py tests/my_clip.wav
```

Adding New Audio Clips

```powershell
# Convert .ogg to .wav (change batchX to your batch name)
Get-ChildItem data\raw_voice_notes\pos_batchX\*.ogg | ForEach-Object {
    ffmpeg -i $_.FullName -ar 16000 -ac 1 "data\raw_voice_notes\pos_batchX\$($_.BaseName).wav"
}

# Copy positive clips (change $i to the next number after your last file)
$i = 40
```

```
Get-ChildItem data\raw_voice_notes\pos_batchX\*.wav | ForEach-Object {
    Copy-Item $_.FullName -Destination ("data\dataset\positive\wake_{0:D3}.wav" -f $i)
    $i++
}

# Copy negative clips (change $i accordingly)
$i = 67
Get-ChildItem data\raw_voice_notes\neg_batchX\*.wav | ForEach-Object {
    Copy-Item $_.FullName -Destination ("data\dataset\negative\neg_{0:D3}.wav" -f $i)
    $i++
}
```

> [!IMPORTANT]
> **The single most important thing is DATA.** Every code upgrade in the world cannot compensate for only having 39 positive samples. If you follow this roadmap and reach 150+ samples of each class, your model WILL work reliably.